REPUBLIQUE TUNISIENNE
Ministère de l'Enseignement Supérieur
et de la Recherche Scientifique
Concours Nationaux d'Entrée aux Cycles de
Formation d'Ingénieurs
Session 2016

الجمهورية التونسية
وزارة التعليم العالي
والبحث العلمي

المناظرات الوطنية للدخول
إلى مراحل تكوين المهندسين
دورة 2016

Concours Toutes Options
Epreuve d'Anglais

Date : Jeudi 26 Mai 2016 **Heure : 14 H** **Durée : 2 H** **Nb pages : 08**

Barème : Part I : 30, Part II : 30, Part III : 20

IMPORTANT :

1- *L'épreuve d'anglais comporte deux séries de feuilles :*

- Les énoncés s'étalant sur 4 pages que les candidats sont appelés à garder

- Les feuilles réservées aux réponses (Answers sheets) s'étalant sur 4 pages, lesquelles

Doivent être rendues à la fin de l'épreuve aux professeurs surveillants

2- *Il sera tenu compte de la présentation, (l'écriture au crayon n'étant pas permise)*

Reading passage:

1- The Earth's climate has changed. After nearly two centuries of fossil fuel-burning, concentrations of carbon dioxide in the atmosphere have reached 400 parts per million, especially boosted by the seemingly ever-accelerating amount of combustion in the last few decades according to the World Meteorological Organization. Atmospheric CO2 concentrations of 0.04 percent may not seem like much but it is enough to have already raised average global temperatures by a full degree Celsius, according to the U.K.'s Met Office, with more warming on the way as the greenhouse gas lingers invisibly in the atmosphere, trapping heat, or mixing into the ocean, rendering its waters more acidic.

2- In fact, the world has not seen CO2 concentrations this high in at least hundreds of thousands of years. Roughly 35 billion metric tons of CO2 are spewed into the atmosphere annually—and rising. The waters of the global ocean have become 30 percent more acidic in the last few decades and the world has not been this warm in thousands of years. This year is likely to be the hottest one since record keeping began, thanks to an El Nino weather pattern that's taking place in addition to global warming. The top 10 warmest years have all occurred since 1998, which was the year of the last major El Nino.

3- Worse, farming, forest-clearing and other activities have contributed to emissions of other greenhouse gases, such as methane and nitrous oxide, the latter more commonly known as laughing gas, **which is no laughing matter in the atmosphere.**

4-Yet, there are signs of hope as well. The U.S. is burning less coal; Europe, and even China have begun to use less of this dirtiest of fossil fuels. And although India and the rest of Asia are building hundreds of coal-fired power plants, there are also plans for more electricity derived from the sun in India, wind in China

and hot rocks in Indonesia. In fact, renewables are growing fast all over the world, helping keep more CO2 out of the air. Half of the world's electricity could come from less climate polluting sources by 2040, according to the International Energy Agency's latest World Energy Outlook report. The electric output of renewables alone in 2040 may match the electric output of fossil fuel-fired power plants in China, the European Union and the U.S. today.

5-Already, China, the European Union, and U.S.—the world's largest polluters, together responsible for more than half of global pollution—have agreed to limit future greenhouse gas emissions. Compared with 1997 when the Kyoto Protocol to combat climate change was agreed on or 2009 when another effort to craft **a global deal collapsed in Copenhagen**, the prospects for a global effort to combat climate change have never been better. With climate negotiations under way in Paris now, there is a real chance for a comprehensive set of actions from more than 190 nations around the world, all to restrain global warming.

6-There are yet more signs of hope, like the fact that cities, states, provinces and even nations have started to put a price on CO2 pollution in a bid to cut back its use. We also now live in a world where an infrastructure project like the Keystone XL Pipeline to connect Alberta's tar sands with Texas's heavy oil refineries can be rejected because of its perceived impact on climate change. In other words, what once seemed impossible—stopping a fossil fuel project in a world that derives nearly 90 percent of its energy from fossil fuels—has become not only possible, but reality.

7-There is still a long way to go, as that 90 percent figure implies. The gap between what nations have promised—reductions of as much as 11 billion metric tons of greenhouse gases by 2030—and what is needed to avoid adding another degree C to the global average temperature—an additional reduction of at least seven billion metric tons of CO2 by 2030—remains large. There are thousands of fossil fuel-fired power plants, more than a billion petroleum-powered vehicles and all the attendant infrastructure to service them, whether oil pipelines, roads or coal mines. We may need technologies like CO2 capture and storage, even if only to deal with the climate-changing pollution from natural gas-fired power plants or the cement kilns and steel foundries needed to build wind turbines or nuclear reactors.

8-We have entered what might be called the Anthropocene thermal maximum, an era of global warming driven by one species penchant for burning fossil fuels and cutting down forests. Right now in 2015 may be the last time anyone breathes air with average CO2 concentrations below 400 ppm, as this number marches seemingly inexorably upward. But we don't have to keep adding to that number forever.

By David Biello on November 10, 2015

Part A- READING COMPREHENSION (30 marks)

I- Choose the best alternative. The text conveys the idea that (1 mark):

*1- Global warming is not as hazardous as we present it.

*2- Global warming has reached alarming levels that's why urgent action is imperative.

*3- Fossil fuels have to be banned altogether by international laws to save our planet.

II- Referring to the text, complete the table with the appropriate information about global warming (6 marks)

III- Complete the following sentences with reference to the text (3 marks):

*1- Meteorologists noticed that..... **(a)** .....were the hottest ever.

*2- It is expected that some countries will resort to..... **(a)** ..... in an attempt to ..... **(b)** .....

IV- Select the correct option (3 marks):

*1- Scientists think the globe is heading toward:

(a) a warmer age/ **(b)** devastating cyclones/ **(c)** total destruction

*2- El Nino refers to:

(a) a renewable source of power / **(b)** a new greenhouse gas / **(c)** a warm ocean current

*3- Greenhouse gases can:

(a) combat climate change / **(b)** not be seen in the air/ **(c)** be caused by El Nino

V- Answer the following questions :

1-Despite the alarming situation, the writer seems somewhat optimistic about the possibility to reduce global warming in order to limit its negative effects.

Give 3 expressions to justify **(3marks)**

2- According to the writer, the measures taken by the industrialized countries are not sufficient to restrain global warming. Pick out 2 justifications from the text **(2 marks)**

VI- Say whether the following statements are True or False. Justify from the text (4 marks)

*1- Renewables could never be an alternative to fossil fuels in the generation of electricity.

*2- Harming the atmosphere is an act to be paid for to preserve a respectable quality of the air

VII- Explain the following expressions (2 marks):

*1- Which is no laughing matter in the atmosphere (§3)

*2- The global deal collapsed in Copenhagen (§5)

VIII- What do the underlined words in the text refer to? (3 marks)

*1- this (§ 2) *2- this dirtiest of fossil fuels (§4) *3- its (§6)

IX- Find words in the text that have approximately the same meaning as (3marks):

1- about(§2) 2- factories for producing energy (§7) 3- period of time (§8)

Part B- LANGUAGE (30 marks)

I- Put the bracketed words in the right tense or form (10 marks)

This morning, a Muslim man **[1] (throw off)** a National Express coach for 'looking shifty'. Due to this ordeal, the 42-year-old, who **[2] (currently/ study)** Chinese at the Open University, **[3] (feel)** angry, discriminated against and treated differently to everyone else because of the way he **[4] (dress)** and the fact he's got a beard.

[5] (Origin) from Somalia, Mr. Ismail – a practicing Muslim – **[6] (flee)** his worn torn country 15 years ago and **[7] (live)** in the UK as a British citizen ever since. He believes the other passengers became **[8] (suspect)** when they saw that he **[9] (carry)** three bags of luggage.

A spokesman for National Express said they **[10] (investigate)** the matter to determine liability.

II- Complete with the appropriate word. (The first two letters the word begins with and a synonym of the word are provided to help you) (10 marks)

Cannabis can **[1] re... (= ease)** constant pain more **[2] ef... (= efficiently)** than conventional medicines, according to a new study likely to flame the **[3] de... (= discussion)** on the medical use of marijuana. Scientists from the Australian National Drug and Alcohol Centre **[4] di... (= found**

out) that patients with **[5] ch... (= constant)** pain who used the drug said it diminished their symptoms better than opioid medications, which are **[6] hi... (= extremely)** addictive and can cause accidental overdoses. The **[7] re... (= study)** analyzed 1,500 patients, aged in their late 40s and early 50s, who suffered from conditions including back **[8] ac... (= pain)**, migraines and arthritis, and whose **[9] ph... (= doctors)** prescribed them heavy-duty opioid medications, such as morphine and oxycodone. Professor Louisa Degenhardt, from the National Drug and Alcohol Centre and the University of Melbourne, **[10] le... (= conducted)** the study. Her team found that almost 13 per cent of the patients had used the illegal drug in the past year on top of their prescribed medication.

III- Express differently starting as given on the answer sheet (keep the same meaning) (10 marks)

1- I am sure it was my grandmother who cooked the lunch.

2- Providing you return the product, we will refund you.

3- He did not accept the offer because the salary was low.

4- They like the vegetables they grow at home.

5- "I'm sorry I broke the vase, Jane".

6- The economic crisis occurred because the government didn't take strict measures.

7- Neither Steve nor Paul watched the movie.

8- We think the thief robbed the bank last night.

9- I prefer staying at home to going out.

10- Despite his extreme wealth, he does not help the poor.

Part C- WRITING (20 marks)

I- Correct the underlined mistakes (2.5 marks):

*1- : They are **hold** responsible for this terrorist act.

*2-: **Due to** his wealth, he can afford to buy this luxurious palace.

*3-: So far, we **are doing** three exercises.

*4-: Read the question carefully not to **do** any mistakes.

*5-: I advise you to **try** on your coat because it's cold outside.

II- Re-order to obtain a coherent paragraph (2.5 marks):

*a- Save the world before it's too late.

*b- are felt across the globe,

*c- As the impacts of a warming climate

*d- with a mission as simple as it is daunting:

*e- our leaders are gathering in Paris

III- Essay (15 marks)

According to you, is nuclear energy- as an alternative source of power to fossil fuels - worth the risks? Use solid convincing arguments to support your point of view in a 15-line essay.

Suggested Answers

Part A- READING COMPREHENSION :

I- Choose the best alternative. The text conveys the idea that (1 mark):

2- global warming has reached alarming levels that's why urgent action is imperative

II- Complete the table with the appropriate information about <u>global warming</u> (6 marks)

<u>Definition</u>	global rise in the earth's temperature (caused by high levels of carbon dioxide and other gases in the atmosphere)
<u>Causes</u>	*fossil fuels burning * high concentrations of CO2 in the air * farming (use of pesticides and all kinds of chemicals)* deforestation* activities emitting greenhouse gases (such as methane and nitrous oxide)
<u>Effects</u>	*warming of the surface of the earth * trapping heat *rendering waters warmer and more acidic * climate change* breathing contaminated harmful air etc...
<u>Solutions</u>	*limit future greenhouse gas emissions (decrease the use of fossil fuels) * resort to renewables or nuclear energy *stop fossil fuel projects* penalize great polluters* develop new technologies to be able to deal with CO2 etc...

NB: If the student gives three correct items or more give full mark (1.5pts for Causes/ Effects/ Solutions)

III- Complete the sentences with reference to the text (3 marks):

*1-Meteorologists noticed that **the last two decades (= the last years of the 20th century and beginning of 21st)** were the hottest ever

*2-It is expected that some countries will resort to **renewables/ put a price on CO2 pollution** in an attempt to **reduce global warming.**

IV- Select the correct option (3 marks):

*1- Scientists think that the globe is heading toward: ***(a)*a warmer age**

*2- *El Nino* refers to: ***(c)*a warm ocean current**

*3-Greenhouse gases can: ***(b)*not be seen in the air**

V- Answer the following questions :

1- Three <u>expressions</u> to justify **(3marks)**

***a-: yet, there are <u>signs of hope</u> /*b-there are yet <u>more signs of hope</u> / *c-:prospects ... <u>have never been better</u>/*d-: there is <u>a real chance</u>/ *e-: has become <u>not only possible, but reality</u>**

2- Two justifications

***a-: there is still a long way to go / * b-: the gap... remains large / * c-: ...may be the last time anyone breathes air with average concentrations below 400 ppm / *d-: we don't have to keep adding to that...**

VI- True or False. Justify from the text (4 marks)

*1-(FALSE) : Half of the world's electricity could come from less climate polluting sources by 2040/ the electric output of renewables alone in 2040 may match..... and the U.S today.

*2-(TRUE) : cities, states, provinces and even nations have started to put a price on CO2 pollution in a bid to cut back its use.

VII- VII- Explain the expressions (2 marks):

*1-: unlike its name, the damage occurring in the atmosphere because of this gas is not something funny or amusing but really serious and alarming

*2-: the participating nations did not reach an agreement to be implemented by all

VIII- What do the underlined words in the text refer to? (3 marks)

*1-: CO2 concentrations of 0.04 percent = CO2 concentrations having reached 400 parts per million *2-: coal *3: infrastructure project

IX- Find words in the text that have approximately the same meaning (3marks):

1- roughly 2- (power) plants 3- era

Part B- LANGUAGE :

I- Put the bracketed words in the right tense or form (10 marks)

1	was thrown off	6	fled
2	is currently studying	7	has been living/ has lived
3	felt	8	suspicious
4	was dressed	9	was carrying
5	originally	10	would investigate

II- Complete with the appropriate word (10 marks)

1	relieve	6	highly
2	ef fectively	7	research
3	debate	8	ache
4	discovered	9	physicians
5	chronic	10	led

III- Express differently starting as given on the answer sheet (keep the same meaning) (10 marks)

1- *My grandmother must* **have cooked the lunch.**

2- *Unless* **you return the product, we will not refund you.**

3- *He turned* **down the offer because the salary was low.**

4- *They like the* **home-grown** *vegetables.*

5- *John apologized to Jane* **for breaking the vase (=for having broken the vase).**

6- *Had* **the government taken strict measures, the economic crisis wouldn't have occurred.**

7- *Steve didn't watch the movie and neither* **did Paul.**

8- *The thief is thought* **to have robbed the bank last night.**

9- *I'd rather* **stay at home than go out.**

10- *Although* **he is extremely wealthy, he does not help the poor.**

Part C- WRITING :

I- Correct the underlined mistakes (2.5 marks):

*1- : They are **hold** responsible for this terrorist act → (**held**)

*2-: **Due to** his wealth, he can afford to buy this luxurious palace→ (**thanks to/ owing to**)

*3-: So far, we **are doing** three exercises → (**have done**)

*4-: Read the question carefully not to **do** any mistakes → (**make**)

*5-: I advise you to **try** on your coat because it's cold outside → (**put**)

II- Re-order to obtain a coherent paragraph (2.5 marks):

As-are-our-with-Save → (**c-b-e-d-a**) → **n° (3-2-5-4-1)**

(**c**) As the impacts of a warming climate (**b**) are felt across the globe, (**e**) our leaders are gathering in Paris (**d**) with a mission as simple as it is daunting: (**a**) Save the world before it's too late.

III- Essay (15 marks)

Suggested marking scheme: 5 pts: ideas / 2 pts: grammar / 2pts: vocabulary / 2pts: spelling / 2 pts: cohesion & coherence/ 2pts: punctuation